

Trajan Hammonds

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Research Interests	Analysis of automorphic forms, asymptotics of special functions, harmonic analysis	
Employment	Massachusetts Institute of Technology , Cambridge, MA NSF Postdoctoral Fellow	2026-2029
	Aarhus University , Aarhus, Denmark Postdoctoral Researcher	2025-2026
Education	Princeton University , Princeton, NJ PhD in Mathematics Advisor: Akshay Venkatesh	2020-2025
	Carnegie Mellon University , Pittsburgh, PA M.S. in Mathematics B.S. in Mathematics	2016-2020
	Independent University of Moscow , Moscow, Russia Study Abroad	2018
Publications and Preprints	• Trajan Hammonds . Relative Character Asymptotics Beyond Stability for $\mathrm{PGL}_2 \times \mathrm{GL}_1$. preprint, (2026)	
	• Anshul Adve, and Trajan Hammonds . Optimal L^4 Estimates via Fourier Analysis . preprint, (2025)	
	• Trajan Hammonds , Seoyoung Kim, Steven J. Miller, Arjun Nigam, Kyle Onghai, Dishant Saikia, and Lalit Sharma. k-Diophantine m-tuples in Finite Fields . <i>Int. J. Number Theory</i> 19 (2023) No. 4, 891-912	
	• Fatma Cicek, Giuliana Davidoff, Sarah Dijols, Trajan Hammonds , Aaron Pollack, and Manami Roy. The completed standard L-function of modular forms on G_2 . <i>Math. Z.</i> , 302 , (2022), 483-517	
	• Trajan Hammonds , Casimir Kothari, Noah Luntzlara, Steven J. Miller, Jesse Thorner, and Hunter Wieman. Explicit Sato-Tate conjecture for primes in arithmetic progressions . <i>Int. J. Number Theory</i> 17 (2021) No. 8, 1905-1923	
	• Trajan Hammonds , Seoyoung Kim, Benjamin Logsdon, Álvaro Lozano-Robledo and Steven J. Miller. Rank and bias in families of hyperelliptic curves via Nagao's conjecture . <i>Journal of Number Theory</i> , Volume 215, (2020), 339–361.	
	• Trajan Hammonds , Jeremy Johnson, Angela Patini, and Robert Walker. Counting roots of polynomials over $(\mathbb{Z}/p\mathbb{Z})^2$. <i>Houston Journal of Mathematics</i> , Volume 44, Number 4 (2018), 1111–1119	
Skills	• Languages : Python, C, HTML • Libraries and Tools: PyTorch, NumPy, Pandas, SciPy, Matplotlib, PyPortfolioOpt	
Awards	• NSF US Junior Oberwolfach Fellow (2023)	
	• Princeton University President's Fellowship (2020)	
	• AMS Math in Moscow Scholarship (2018)	

Invited Talks	• International Seminar on Automorphic Forms, Online	June 2026
	• London Number Theory Seminar, London, UK	Feb 2026
	• MIT PDE/Analysis Seminar, Cambridge, MA	May 2025
	• Johns Hopkins Junior Number Theory Days, Baltimore, MD	Feb 2025
	• Princeton/IAS Joint Number Theory Seminar, Princeton, NJ	Dec 2024
Conferences and Workshops	• Pittsburgh Links..Analysis and Number Theory, Pittsburgh, PA	March 2024
	• AIM Workshop : ..Automorphic Forms, Pasadena, CA	February 2024
	• CMI Workshop : ..Automorphic Forms, Oxford, UK	September, 2023
	• ArStAFANT Workshop, EPFL, Switzerland	June 2023
	• Oberwolfach Seminar : ...L-functions.. Oberwolfach, Germany	May 2023
	• Arizona Winter School : Unlikely Intersections, Tucson, AZ	March 2023
	• Automorphic Forms Summer School, Erdos Center, Hungary	September 2022
	• Masterclass on Relative Trace Formula, Copenhagen, Denmark	August 2022
	• Summer School on the Langlands Program, IHES, France	July 2022
	• Arizona Winter School : Automorphic Forms Beyond GL ₂ , Tucson, AZ	March 2022
	• ICTS Elliptic Curves and Special Values of L-functions, Virtual	August 2021
	• Hausdorff Summer School : Circle Method, Virtual	May-June 2021
	• Research in Number Theory, Virtual workshop	October 2020
	• UConn Number Theory Summer School, Virtual conference	June 2020
	• Bhargavology Learning Seminar, Stanford, Virtual seminar series	April–May 2020
	• Joint Mathematics Meetings, Denver, CO	January 2020
	• Duluth REU, University of Minnesota Duluth, Duluth, MN	June–August 2019
	Advisor : Joe Gallian	
	• CBMS Conference on L-functions and Multiplicative Number Theory, University of Mississippi, Oxford, MS	May 2019
	• Automorphic Forms Workshop, Duquesne, Pittsburgh, PA	March 2019
	• Joint Mathematics Meetings, Baltimore, MD	January 2019
	• Quebec-Maine Number Theory Conference, Université Laval, Quebec City, Quebec, Canada	October 2018
	• SMALL REU, Williams College, Williamstown, MA	June–August 2018
	Advisors: Seoyoung Kim, Steven J. Miller, and Jesse Thorner	
	• UConn Number Theory Summer School, University of Connecticut, Sorrs, CT	May–June 2018
	• Joint Mathematics Meetings, San Diego, CA	January 2018
	• MSRI Undergraduate Program, MSRI, Berkeley, CA	June–August 2017
	Advisors: J. Maurice Rojas and Federico Ardila	
Teaching	Academic Coordinator, MathROOTS, MIT	July 2026
	• Designed curriculum, wrote problem sets and solutions, gave lectures, completed admissions, oversaw problem sessions, assisted in day-to-day operations, engaged in vertical mentoring.	
	Assistant Academic Coordinator, MathROOTS, MIT	July 2025
	• Gave lectures, wrote problem sets and solutions, wrote and graded exams, oversaw problem sessions, assisted in day-to-day operations, engaged in vertical mentoring.	
	Academic Mentor, MathROOTS, MIT	June-July 2024
	• Gave lectures, oversaw problem sessions, assisted in day-to-day operations, engaged in vertical mentoring.	
	Academic Mentor, MathROOTS, MIT	June-July 2023
	• Gave lectures, oversaw problem sessions, assisted in day-to-day operations, engaged in vertical mentoring.	
	Teaching Assistant, BEAM (Bridge to Enter Advanced Mathematics) Discovery Program, Virtual	July–August 2020

- Led activities, oversaw problem sessions, assisted in day-to-day operations, and co-TA'd the course *Count Without Counting*. Interacted with over fifty 6th graders daily.

Teaching Assistant and Grader, Carnegie Mellon University August 2018–May 2020

- Gave recitations for Concepts of Mathematics (21-128, 4.66/5 Rating), Matrices and Linear Transformations (21-241, 4.52/5 Rating), Graded for Principles of Real Analysis I (21-355)

Course Assistant/Grader, Art of Problem Solving August 2017–March 2019

- Online course assistant for Stretch Algebra, Intro to Number Theory, Precalculus and WOOT (Worldwide Online Olympiad Training)

Activities and Service

- Member of Climate and Inclusion Committee (2023–Present)
- Group leader for Mentoring Möbius, Princeton University (2020–2022)
- Referee for *Journal of Number Theory*, *Essential Number Theory*